LinuxDS 帧率动态调整性能数据报告_Multi150_2022_11_09

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一、测试环境

地图：Multi_150

人数：3p

版本：Dev

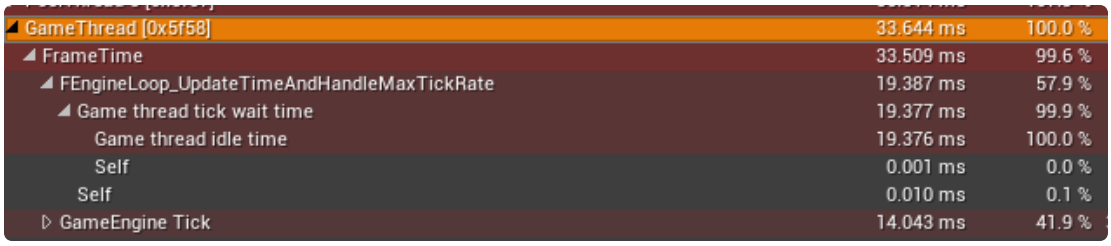
DS 帧率测试：DS 分别以 15 / 20 / 30 帧，获取 stat / llm / networkprofiler 性能数据

二、性能数据分析

1、Stat

（1）数据

30 帧



GameThread [0x5f58]	33.644 ms	100.0 %
FrameTime	33.509 ms	99.6 %
FEngineLoop_UpdateTimeAndHandleMaxTickRate	19.387 ms	57.9 %
Game thread tick wait time	19.377 ms	99.9 %
Game thread idle time	19.376 ms	100.0 %
Self	0.001 ms	0.0 %
Self	0.010 ms	0.1 %
GameEngine Tick	14.043 ms	41.9 %

20 帧

GameThread [0x6d18]	50.120 ms	100.0 %
FrameTime	49.906 ms	99.6 %
FEngineLoop_UpdateTimeAndHandleMaxTickRate	32.614 ms	65.4 %
Game thread tick wait time	32.602 ms	100.0 %
Game thread idle time	32.600 ms	100.0 %
Self	0.002 ms	0.0 %
Self	0.012 ms	0.0 %
GameEngine Tick	17.214 ms	34.5 %

15 帧

GameThread [0x526a]	66.754 ms	100.0 %
FrameTime	66.510 ms	99.6 %
FEngineLoop_UpdateTimeAndHandleMaxTickRate	47.457 ms	71.4 %
Game thread tick wait time	47.444 ms	100.0 %
Game thread idle time	47.443 ms	100.0 %
Self	0.002 ms	0.0 %
Self	0.012 ms	0.0 %
GameEngine Tick	18.969 ms	28.5 %

（2）结论

DS 帧率下降后，CPU 占用率明显大幅减少

[30 帧]　　[20 帧]　　[15 帧]

41.9%　=>　34.5%　=>　28.5%

PS：GameEngineTick Time 正常会有±1~2ms 左右的浮动

2、LLM

（1）数据

30 帧

	A	C
		Total
AVG:		
	2932.3	506.06
MAX:		
	2945.36	571.13

20 帧

A	C
	Total
AVG:	
	560.89
MAX:	
	686.64

15 帧

A	C
	Total
AVG:	
2943.19	548.21
MAX:	
2968.59	649.18

（2）结论

随着帧率下降，llm avg 有微幅上涨

[30 帧]　　[20 帧]　　[15 帧]

506MB　=>　560MB　=>　548MB

↑ 10%　　↑ 8.3%

PS：正常会有±5%左右的浮动

3、NetworkProfiler

（1）数据

30 帧

```
Summary
    Frames              : 25457.0
    Seconds             :   854.4
Details (PER SECOND)
    Outgoing bandwidth :    46.8 KB/s
```

```
Summary
    Frames              : 25457.0
    Seconds             :   854.4
Details (PER SECOND)
Details (TOTAL)
    Outgoing bandwidth :    39.1 MBytes
```

20 帧

```
Summary
    Frames              : 10976.0
    Seconds             :   552.9
Details (PER SECOND)
    Outgoing bandwidth :    56.4 KB/s
```

```
Summary
    Frames              : 10976.0
    Seconds             :   552.9
Details (PER SECOND)
Details (TOTAL)
    Outgoing bandwidth :    30.4 MBytes
```

15 帧

```
Summary
    Frames              :  7827.0
    Seconds             :   526.0
Details (PER SECOND)
    Outgoing bandwidth :    53.2 KB/s
    SocketSend Count   :     0.0 hz
```

```
Summary
    Frames              :  7827.0
    Seconds             :   526.0
Details (PER SECOND)
Details (TOTAL)
    Outgoing bandwidth :    27.3 MBytes
```

（2）结论

随着帧率下降，网络带宽有一定程度的上涨，如果调整 DS 帧率，带宽是需要重点优化

[30 帧]　　[20 帧]　　[15 帧]

46.8KB/s　=>　56.4KB/s　=>　53.2KB/s

↑ 20%　　↑ 13%

PS：正常会有±5% 左右的浮动